

5. Build and Verify

A weekend to build, 48 hours to cure

Measure, inspect, remake, prove

THIS IS A PROJECTILE WEAPON

A combustion launcher throws a half-pound potato several hundred feet at speeds that will kill a person. It is not a toy, it is not a firework, and the fact that it is made of plumbing parts changes nothing about what it does downrange. Every rule on the **Safety** sheet applies every time: adult present and in charge, muzzle discipline as strict as a rifle's, a confirmed clear range, eyes and ears on, and never a person, animal, building, road or anything you cannot afford to destroy in the direction it is pointing.

The standard

Do not reward a finished-looking launcher. Reward a documented sequence of square cuts, clean sockets, fully seated joints, a complete cure, and an adult's final decision. A doubtful pressure part is not repaired or "good enough": mark it REJECT, make it unusable, and replace it.

Procedure: parts and acceptance

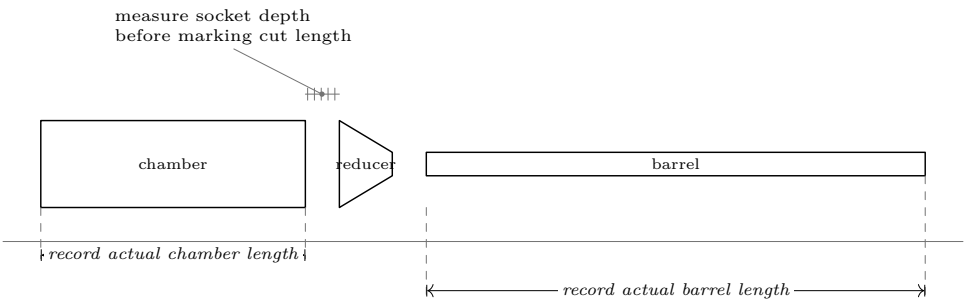
Part	Required evidence	Reject if
Chamber	3 in Sch 40 pressure PVC; rating and manufacturer print legible	DWV, foam/cellular core, unknown rating, crack, whitening, deep gouge
Cleanout	Matching pressure-rated adapter and undamaged threaded plug	Plug binds, rocks, or has damaged threads; plug must never be glued
Reducer	Pressure-rated fitting; sockets clean and round	Socket chipped, distorted, contaminated, or a loose dry fit
Barrel	1.5 in Sch 40 pressure PVC; straight and fully printed	Split, deep score, distortion, or unknown rating
Igniter assembly	Piezo unit; stainless screws, nuts, and washers	Loose hardware, damaged insulation, unreliable spark
Joining materials	Compatible primer and pressure-PVC cement, in date	Gel, skin, contamination, missing label, expired shelf life

Pressure-rated pipe only. Schedule 40 *pressure* PVC, printed with an NSF-PW rating and a pressure figure. Never DWV, never foam-core, never cellular-core, never thin-wall drain pipe — those are not rated for internal pressure at all and they fail by shattering into fragments rather than splitting. This is the one specification in the whole project that is not negotiable.

Before touching a tool

Question	Record	Adult check
Can you point to the pressure rating on every pressure part?	Rating: _____	Initials/date: _____
Which face will be the chamber end of each fitting?	Pencil arrows made: yes / no	All mating faces identified: yes / no
Where will the assembly cure untouched for 48 hours?	Place: _____	Start time: _____

Procedure 1: lay out and measure



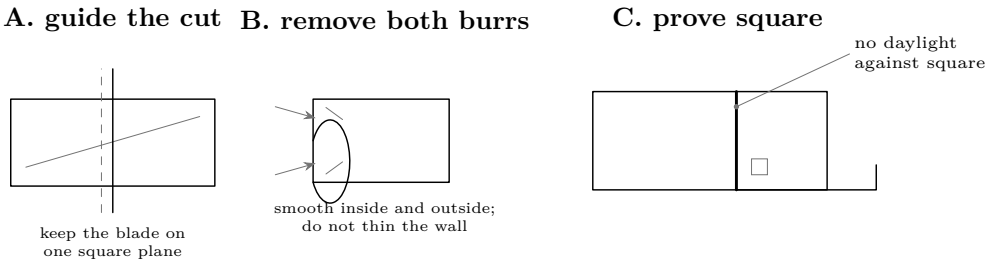
- 1. Place every dry, uncemented part in assembly order.
- 2. Match each printed rating to the parts list. If any mark is absent or does not match, label that part HOLD and replace it.
- 3. Photograph or sketch every rating mark before a fitting can cover it.
- 4. Measure each socket depth with a rule.
- 5. Transfer that depth to its pipe end as an insertion mark.
- 6. Draw one alignment line across each dry-fitted joint.
- 7. Measure each pipe diameter at two points 90 degrees apart. If the readings differ, label the pipe HOLD and replace it.
- 8. Dry-fit one joint without forcing. The pipe must reach its insertion mark and remain snug; otherwise separate it and replace the mismatched part.

Layout record			
Feature	Planned	Measured	Pass / remake
Chamber cut length	_____	_____	_____
Barrel cut length	_____	_____	_____
Reducer socket depth	_____	_____	_____
Cleanout socket depth	_____	_____	_____

Pause and explain. Why does a socket-depth mark reveal an incomplete joint? Why is a square end stronger than a slanted end? Have the builder answer before cutting. *see the 01 Measure and Mark sheet*

Rating-mark evidence		
Part	Mark copied or photo ID	Witness / decision
Chamber		
Barrel		
Reducer and cleanout		

Procedure 2: cut, deburr, and inspect



1. Clamp the pipe firmly without changing its round shape. If the wall flattens, release the clamp and reset it.
2. Cut once on the waste side of the mark with the square guide shown in A.
3. Remove the inner burr around the full circumference.
4. Remove the outer burr around the full circumference.
5. Add only a small chamfer. If the wall is visibly thinned, reject the end.
6. Hold a square at each of four quarter-turn positions as shown in C.
7. Shine light behind the square and record the largest visible gap. If any position is visibly slanted, mark the end HOLD and correct or replace it.
8. Wipe the end with a dry white cloth.
9. Inspect the clean end under raking light. Any chip, crack, deep score, or heat damage sends the part to the remake decision below.

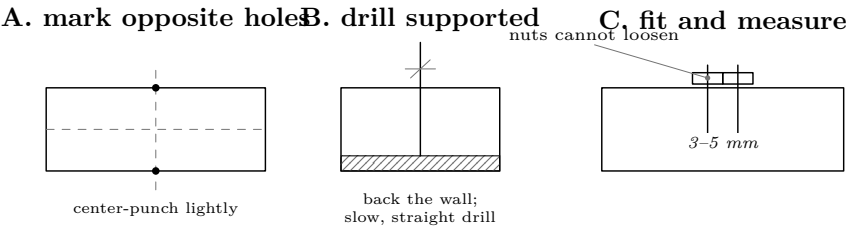
End	Gap	Inner burr	Outer burr	Decision / initials
Chamber A	_____	none / found	none / found	_____
Chamber B	_____	none / found	none / found	_____
Barrel A	_____	none / found	none / found	_____
Barrel B	_____	none / found	none / found	_____

REMAKE DECISION

If an end is visibly slanted, chipped, cracked, deeply scored, heat-damaged, or too short after correction, do not advance it. Mark the defect, cut back to sound material if the planned dimensions still permit, or replace the pipe. Never fill a bad cut with cement or epoxy.

Raking-light evidence		
End	Observation and largest gap	Accept / remake; initials
Chamber ends		
Barrel ends		

Procedure 3: drill and prove the ignition assembly



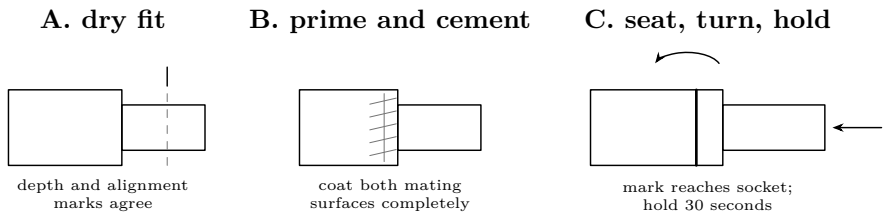
1. Mark two opposite hole centers at the approved chamber location shown in A.
2. Dry-position the hardware outside the wall. If it conflicts with the cleanout, reducer, or hand clearance, erase the marks and return to the drawing.
3. Support the inside wall as shown in B.
4. Drill one small pilot hole at each center.
5. Enlarge each hole only to the fastener diameter.
6. Deburr each hole without enlarging it. Any crack, craze, or oversize hole requires replacement of the chamber.
7. Install the washers and nuts.
8. Measure the closest electrode spacing shown in C. Adjust it to 3–5 mm; if the hardware will not stay immobile, replace the faulty hardware or chamber.
9. Apply only the seal specified on Craft 05.
10. Open the cleanout and verify that the chamber is empty and unfuelled.
11. Have the adult click the piezo ten times while viewing from the open end. Fewer than ten visible sparks is a failed check; stop and correct the assembly.

Ignition inspection		
Check	Measurement / result	Decision
Electrode gap at closest point	_____ mm	pass / remake
Ten empty-chamber clicks	_____ sparks seen	pass / remake
Hardware movement by hand	none / present	pass / remake
Hole edge under raking light	clean / cracked / enlarged	pass / remake

Pause and explain. Why test the ignition before cementing the chamber? What defect would make replacement safer than adding more sealant? *see the 05 Drill and Seal sheet*

Ten-click witness record		
Clicks	Observed spark count / pattern	Adult witness / decision
1–5		
6–10		

Procedure 4: dry-fit, solvent-weld, and witness



1. Rehearse one joint dry: orient, insert to the depth mark, quarter-turn, restore alignment, and hold.
2. Assign one adult to call time.
3. Assign one adult to watch the depth and alignment marks.
4. Confirm that ventilation and conditions meet both product labels. If they do not, stop and move the work or postpone it.
5. Prime both mating surfaces of one joint as shown in B.
6. Cement both mating surfaces according to the label.
7. Push the pipe to its depth mark.
8. Turn it one quarter-turn.
9. Restore the alignment marks.
10. Hold it still for 30 seconds. If it backs out or either mark misses, reject the joint and remake the affected assembly.
11. Wipe only loose exterior excess without moving the joint.
12. Record the fresh bead through a full rotation. A gap, dry area, or uncertainty is a rejection, not a patch instruction.

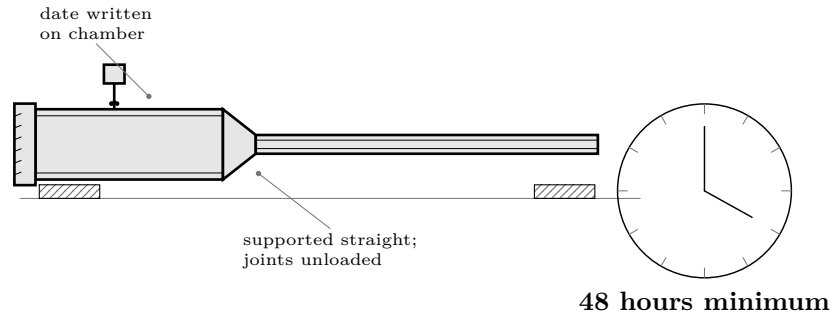
Joint	Start time	Held to	Mark seated	Witness / decision
Cleanout–chamber	_____	_____	yes / no	_____
Chamber–reducer	_____	_____	yes / no	_____
Reducer–barrel	_____	_____	yes / no	_____

STOP, DO NOT PATCH

Reject a joint that backs out, misses its depth mark, rotates away from its alignment mark, contains a visible dry area, or was assembled after the cement had skinned. Do not add cement around the outside. Retire the affected pressure part and remake the assembly. *see the 04 Solvent Weld sheet*

Fresh-joint witness		
Joint	Full-rotation bead observation	Witness / disposition
Cleanout–chamber		
Chamber–reducer		
Reducer–barrel		

Procedure 5: cure and inspect



1. Write the latest joint date and time on the chamber.
2. Calculate the later of 48 hours or the label-required cure.
3. Write that earliest inspection time on the chamber.
4. Place the assembly horizontally on the two broad supports shown.
5. Leave it untouched until the written time. Any movement or uncertain cure history resets the hold pending the adult's label-based decision.
6. After cure, rotate each joint once under bright raking light.
7. Record shoulder seating, mark alignment, and bead continuity. Any gap, crack, whitening, bulge, or disturbed mark rejects the affected assembly.
8. Turn the cleanout plug fully home by hand on dry threads. Binding, rocking, or cross-threading rejects the cleanout; never force or glue the plug.

Cure record

Evidence	Record	Adult initials
Assembly finished	date/time: _____	_____
Earliest allowed inspection	date/time: _____	_____
Actual inspection	date/time: _____	_____
Label required longer cure?	no / yes: _____	_____

360-degree inspection

Feature	Result	Defect location / decision
Cleanout joint and threads	pass / reject	_____
Chamber surface and electrodes	pass / reject	_____
Reducer joints	pass / reject	_____
Barrel surface and muzzle	pass / reject	_____

Cure arithmetic. Label minimum _____ hours; project minimum 48 hours.

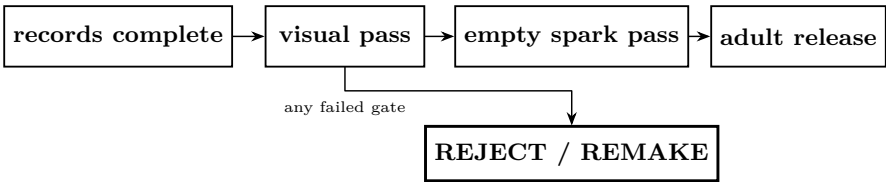
Later governing time: _____ Actual elapsed time: _____ Checked by: _____

Inspection notes

Area	Lighting, angle, and observation	Decision
Threads and plug		
Joints and wall		

Record packet: final proof and release

The procedure is complete only when the earlier bench records and every gate below are complete. Consolidate the detached craft sheets and the build log behind this page; this final record is their index and release decision.



Gate	Result	Evidence
Every pressure part was identified and accepted before work	pass / reject	Parts record complete
Every cut and joint has a measurement and witness decision	pass / reject	Pages 2–5 complete
At least 48 hours and the label-required cure elapsed	pass / reject	Cure times complete
Full surface and all joints passed bright-light inspection	pass / reject	No defect left blank
Cleanout operates by hand; electrodes are immobile	pass / reject	Adult observed
Ten empty, unfuelled ignition clicks produced ten clear sparks	pass / reject	Count: ____/10
Safety sheet was read aloud at the chosen legal range	pass / reject	Reader: _____

NO IMPROVISED PRESSURE PROOF

Do not plug the barrel, add compressed air, hydrostatically pressurize this assembly, strike it, heat it, or perform a close-range “test shot.” Those tests introduce hazards and do not convert questionable workmanship into proof. A failed inspection means retire and remake.

Adult release

I reviewed the complete record, inspected the assembly after cure, confirmed that every gate above passes, and accept responsibility for deciding whether and where any later operation is lawful and safe.

Adult name: _____ Signature: _____ Date: _____

Operation is a separate gate

This build record gives no permission to fire. Before any use, the supervising adult must apply the **Safety** sheet, local law, landowner rules, weather, backstop, exclusion zone, eye and hearing protection, muzzle control, and misfire procedure. Use only the single approved operating procedure; do not vary fuel, projectile fit, barrel, angle, or other variables to seek more pressure, speed, or range. If performance seems abnormal, stop and inspect; never “tune” a combustion launcher.

Final question. What evidence would make you retire this launcher today? Write three observations before any operation: _____, _____, and _____.

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